A degenerated-continuum Ritz formulation for the free vibration of rotating pre-twisted, tapered and cambered NACA compressor blades

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Abstract

A semi-analytical framework is developed for the free vibration and dynamic stability analysis of rotating compressor blades modelled as pre-twisted, tapered, and cambered NACA four-digit airfoil shells. The blade geometry is described using an exact parametric mid-surface mapping with analytically evaluated derivatives, while three-dimensional strains are obtained from the displacement gradient using the inverse mapping Jacobian. A five-field first-order shear deformation model is discretized using the Rayleigh–Ritz method with Jacobi and Legendre polynomial basis functions. Singularities associated with the NACA leading-edge thickness and discontinuities in camber curvature are addressed through specialized coordinate transformations and domain subdivision.

Rotational effects are incorporated through centrifugal prestress, geometric stiffness, Coriolis coupling, and spin-softening. The resulting quadratic eigenvalue problem is solved using an energy-scaled formulation to improve numerical stability and capture divergence-type instabilities. The framework is validated against three-dimensional finite element analyses (ANSYS) and published benchmarks.

The integrated blade mass and volume agree with three-dimensional CAD models within 0.22%. A (6×6) Ritz order (180 degrees of freedom) predicts natural frequencies within 0.07% for the fundamental mode and 0.88% across the primary modes at rotational speeds up to 4,000 rpm. At speeds up to 20,000 rpm, the maximum frequency error remains below 0.84%. Comparisons with published benchmarks yield an RMSE below 0.43%, compared with 1.88% for existing formulations. The proposed framework provides an accurate and computationally efficient approach to predicting the vibration characteristics and dynamic stability of rotating compressor blades.

Keywords: *rotating blade; pre-twist; degenerated shell; first-order shear deformation theory; Rayleigh–Ritz; Jacobi polynomials; centrifugal stiffening; Coriolis effect; gyroscopic eigenvalue problem; dynamic stability; NACA airfoil.*

1. Introduction

1.1. Background and motivation

Compressor and turbine blades operate in a rotating reference frame, where centrifugal forces increase structural stiffness and Coriolis acceleration couples motion in different directions. These effects give the equations of motion a gyroscopic structure. Accurately predicting rotating natural frequencies and tracking their variation with rotational speed on a Campbell diagram [1] are essential for identifying engine-order resonances and reducing the risk of high-cycle fatigue [2,3]. The vibration of rotating turbomachinery blades has been studied for many decades. Southwell and Gough [4] provided early estimates of centrifugal stiffening in airscrew blades, while Carnegie [5] developed variational energy expressions for pre-twisted rotating cantilever blades. Later beam models incorporated initial pre-twist, section taper, and coupled bending–torsion motion using different analytical approaches [6,7]. Broader reviews of rotating beam and blade dynamics are also available [8].

Beam theories have limitations when applied to blades with low aspect ratios, significant chordwise flexibility, or geometric features such as camber and chordwise thickness variations that influence their vibration characteristics. These limitations have motivated the use of plate and shell formulations for blade vibration analysis. Dokainish and Rawtani [9] analyzed rotating cantilever plates using the finite element method, while Ramamurti and Kielb [10] investigated twisted rotating plates and Yoo and Pierre [11] determined the modal characteristics of rotating rectangular plates. Leissa and co-workers applied shallow shell theory within a Ritz framework to analyze blades with

camber and twist [12,13]. Qatu and Leissa [14] investigated the natural frequencies of cantilevered, doubly curved laminated shallow shells.

Shell models were later developed to study pre-twisted rotating blades. Hu et al. [15] analyzed the fundamental vibration of pre-twisted cantilever blades, while Kee and Kim [16] developed a model of initially twisted rotating shells that accounts for centrifugal and Coriolis acceleration. Sinha and Turner [17] treated centrifugal forces as a quasi-static preload on a thin-shell blade. Sun et al. [18,19] developed a general shell model and extended it to arbitrary stagger angles. Sinha and Zylka [20] incorporated airfoil geometry by representing a twisted blade as a thin shell with an equivalent camber radius. Yao et al. [21] combined first-order shear deformation theory (FSDT) with the Rayleigh–Ritz method to analyze blades with NACA four-digit airfoil profiles. Ritz and Chebyshev–Ritz methods have also been applied to sandwich, functionally graded, and coated composite blades [22–25]. Broader reviews of shell vibration analysis are available in the literature [26].

1.2. Literature review and research gaps

A review of the existing literature highlights several issues related to blade geometry.

Geometry. Many of the shell models discussed above formulate the strain energy using classical two-dimensional shell theory [27], in which the surface geometry is described through Lamé parameters and principal curvatures. These quantities can be expressed in closed form for cylindrical, conical, and shallow shell surfaces. However, describing a twisted, tapered, and cambered airfoil with nonorthogonal parametric lines and thickness variations in both directions is more difficult. As a result, some studies have simplified the blade geometry by using a twisted shallow shell [24], a cylindrical panel [25], or an equivalent camber radius [20].

Other studies have addressed some of these geometric limitations. Lotfan and Bediz [28] developed an FSDT shell model with spectral Chebyshev discretization for rotating pre-twisted blades with stagger. Lotfan and Cigeroglu [29] extended this approach to a surface-based framework that accounts for preset angle, pretwist, spanwise and chordwise section variations, and chord taper. Quasi-three-dimensional and three-dimensional models have also been developed for rotating pre-twisted blades and shell panels [30,31].

Methods beyond the Ritz framework have also been used to represent complex blade geometries. These include degenerated-continuum shell elements [32–34], isogeometric analysis, which Hughes et al. [35] introduced and Zhong et al. [36] applied to rotating shell blades with variable thickness, and the Carrera unified formulation [37]. Filiz et al. [38] and Bediz and Aksoy [39] developed spectral Chebyshev solutions using coordinate transformations for three-dimensional pre-twisted and curved beams. The latter study also considered a curved, pre-twisted beam with an airfoil cross-section. Pinho et al. [40] recently proposed a Ritz framework that automatically evaluates shell geometry from the position vector using Büchter–Ramm–Roehl shell theory [34]. Their examples focused on linear free-vibration problems in regular domains.

Rotation and prestress. In rotating shell models, centrifugal stiffening is commonly included by treating centrifugal forces as a quasi-static load [17], integrating the centrifugal forces directly, or performing a prestressed analysis [29]. Lotfan and Bediz [28] showed that including geometric nonlinearity in the static centrifugal analysis can noticeably reduce the predicted natural frequencies. They also found that the linear approach is reliable only at low rotational speeds. These findings highlight the importance of how centrifugal prestress is calculated and incorporated into the subsequent vibration analysis.

The gyroscopic eigenproblem. Many Ritz-based blade studies focus on natural frequency curves, mode veering, and Campbell diagrams [22,25]. The quadratic eigenvalue problem arising from gyroscopic effects has a well-established mathematical structure that can be exploited in numerical solutions [41–43]. However, the extent to which this structure is used in blade vibration models varies across the literature.

NACA sections and Ritz numerics. The NACA four-digit airfoil thickness distribution has a square-root behavior near the leading edge, while the camber line consists of two parabolic segments with a discontinuity in curvature at the point of maximum camber [44]. These features produce non-smooth integrands when a global polynomial basis is used. As a result, the rapid convergence generally associated with orthogonal-polynomial Ritz methods [45–48] may be reduced unless these features are handled appropriately. Although this is a numerical rather than a modeling issue, it affects how many degrees of freedom are needed to obtain accurate results.

Validation data. Experimental data for non-rotating, flat, twisted plates are available from the NASA joint study [49–52]. The study reported considerable differences among the theoretical methods available at the time [52]. Experimental data for rotating blades are more limited, with some studies restricted to low rotational speeds, as noted by Liang et al. [53]. We could not identify published measurements of natural frequency as a function of rotational speed for a cambered, tapered, pre-twisted airfoil blade. Validation therefore commonly relies on comparisons with three-dimensional finite-element models and published theoretical results [22,28], as adopted in the present study.

1.3. Present work

The present study does not claim to be the first to apply Ritz-type methods to rotating NACA shell blades [21] or to represent a blade using a general surface description [29,40]. Rather, based on the literature available to us, we did not find a single formulation that combines the following features:

i. A degenerated-continuum representation of the blade, with the three-dimensional Jacobian derived from the analytical derivatives of the mid-surface position vector and its normal.

ii. A single global Ritz basis applied over the entire blade.

iii. Centrifugal prestress obtained from a three-dimensional static solution.

iv. Coriolis coupling in the dynamic formulation.

v. Treatment of the NACA leading-edge and camber singularities in the numerical quadrature.

The blade is modeled as a degenerated three-dimensional continuum. The strain components are obtained from the Cartesian displacement gradient, transformed into a local orthonormal frame, with the transverse normal strain eliminated using the plane-stress condition. The kinematics follow five-field first-order shear deformation theory (FSDT) [54,55], with three global translations of the mid-surface and two director rotations defined in the local frame. These fields are expanded over the entire blade using L^2-orthonormal Jacobi–Legendre tensor-product basis functions. The spanwise basis functions vanish at the root, enforcing the clamped boundary condition exactly.

The centrifugal body force is included in a three-dimensional static equilibrium analysis. The resulting Cauchy stress at each volume quadrature point is then used to construct the initial-stress stiffness matrix, while the Coriolis acceleration contributes to the skew-symmetric gyroscopic matrix. The resulting quadratic eigenvalue problem is solved using an energy-scaled formulation, and each eigenpair is checked by evaluating its residual. The main contributions of this study are as follows:

i. We develop a Ritz model for pre-twisted, tapered, and cambered NACA blades in which the geometry is defined by the mid-surface position vector and its analytical derivatives, without assuming a shallow shell or twisted plate.

ii. We combine Jacobi–Legendre basis functions that enforce the clamped root condition exactly with a singularity-aware volume quadrature. This includes an $\eta = t^2$ transformation and a domain split at the point of maximum camber to address the leading-edge thickness gradient and the discontinuity in camber curvature.

iii. We develop a prestress-consistent formulation for rotating blades. The three-dimensional Cauchy stress obtained from the static solution is used to construct the initial-stress stiffness matrix, with the same kinematic operators and Jacobian used in the elastic stiffness formulation. The gyroscopic eigenvalue problem is then solved using an energy-scaled eigensolver.

iv. We validate the model against three-dimensional ANSYS solid-element results and published benchmarks, including convergence and mass checks.

The remainder of this paper is organized as follows. Section 2 describes the differential geometry and mapping of the pre-twisted, tapered, and cambered NACA 6412 reference blade. Section 3 presents the FSDT shell kinematics, Rayleigh–Ritz basis functions, rotating energy functionals, singularity-aware volume quadrature, and energy-scaled gyroscopic eigensolver. Section 4 describes the computational implementation, vectorized matrix assembly, and baseline blade configuration. Section 5 presents the geometric verification, convergence study, validation against 3D ANSYS models for non-rotating and rotating cases, and Campbell diagram analysis. Finally, Section 6 summarizes the main findings and conclusions.

2. Blade Geometry and Differential Mapping

2.1. Reference Coordinate Systems and Frames

A right-handed global Cartesian frame (X, Y, Z) with unit vectors $\mathbf{e}_X, \mathbf{e}_Y, \mathbf{e}_Z$ is fixed to the rotor and rotates with it at constant angular velocity $\boldsymbol{\Omega} = \Omega\,\mathbf{e}_Y$, $\Omega = 2\pi N/60$, where N is the rotor speed in rpm. The blade span is directed along X (radial direction), starting at the hub radius R_h. The stacking (pre-twist) axis is the radial line $Y = Z = 0$. The mid-surface is parametrized by the span coordinate $x \in [0, L]$ and the dimensionless chord coordinate $\eta \in [0,1]$, measured from the leading edge ($\eta = 0$) to the trailing edge ($\eta = 1$). The through-thickness coordinate is $\zeta \in [-h/2\,,\, h/2]$.

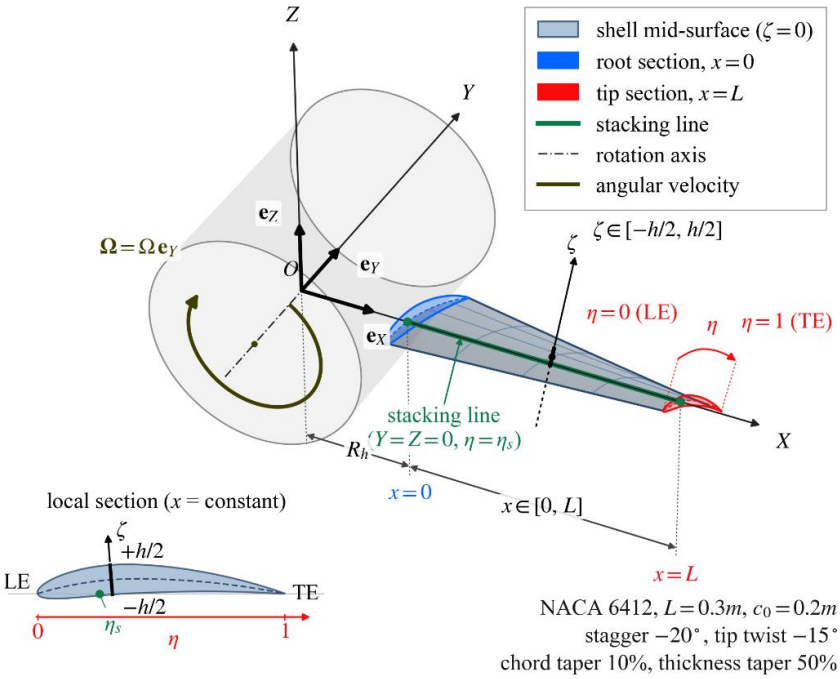


Figure 1. Rotating coordinate systems and mid-surface geometric mapping: Global rotating reference frame (X, Y, Z) with angular velocity $\boldsymbol{\Omega} = \Omega\mathbf{e}_Y$, radial hub radius R_h, and local shell coordinates (x, η, ζ) defined on the pre-twisted compressor blade mid-surface.

2.2. NACA Airfoil Geometry and Camber Discontinuity

The section profile is a NACA four-digit airfoil [59, 60] specified by maximum camber m (first digit, percent of chord), camber position p (second digit, tenths of chord), and maximum thickness-to-chord ratio τ (last two digits, percent of chord). The camber line $f(\eta)$ and its second derivative $f''(\eta)$ are given by:

$$f(\eta) = \begin{cases} \dfrac{m}{p^2}(2p\eta - \eta^2), & \eta \le p \\ \dfrac{m}{(1-p)^2}(1 - 2p + 2p\eta - \eta^2), & \eta > p \end{cases}, \qquad f''(\eta) = \begin{cases} -\dfrac{2m}{p^2}, & \eta \le p \\ -\dfrac{2m}{(1-p)^2}, & \eta > p \end{cases}, \tag{1}$$

so that $f(\eta)$and $f'(\eta)$are continuous across the chord, whereas $f''(\eta)$ exhibits a finite jump discontinuity at $\eta = p$. The section thickness distribution $h_0(\eta)$ is defined as:

$$h_0(\eta) = 10\,\tau\,c\left(0.2969\sqrt{\eta} - 0.1260\,\eta - 0.3516\,\eta^2 + 0.2843\,\eta^3 + a_4\,\eta^4\right), \tag{2}$$

with $a_4 = -0.1015$ for the standard finite trailing-edge thickness and $a_4 = -0.1036$ for a closed trailing edge. The $\sqrt{\eta}$ term produces the rounded leading edge and introduces an integrable singularity in $\partial h_0/\partial\eta$ at $\eta = 0$.

2.3. Spanwise Taper, Pre-twist, and Stacking Axis

The section chord tapers linearly along the span according to $c(x) = c_0(1 - \tau_c x/L)$, while the thickness is scaled independently as $h(x,\eta) = h_0(x,\eta)(1 - \tau_h x/L)$, where c_0 is the root chord length, and $\tau_c, \tau_h \in [0,1)$ denote the non-dimensional chord and thickness taper fractions, respectively.

Each cross-section is rotated about the spanwise stacking axis by the linearly varying angle:

$$\theta(x) = \theta_0 + \kappa x, \qquad \kappa = \frac{\theta_{tip}}{L}, \tag{3}$$

where θ_0 is the root stagger angle, θ_{tip} is the total relative tip pre-twist, and κ represents the constant spanwise twist rate. The corresponding section-fixed unit vectors $\mathbf{e}_y(x)$ and $\mathbf{e}_z(x)$ in the rotating frame are given by:

$$\mathbf{e_y}(x) = \begin{bmatrix} 0 \\ \cos\theta(x) \\ \sin\theta(x) \end{bmatrix}, \qquad \mathbf{e_z}(x) = \begin{bmatrix} 0 \\ -\sin\theta(x) \\ \cos\theta(x) \end{bmatrix}. \tag{4}$$

The radial stacking axis pierces each section on its camber line at the non-dimensional chord fraction η_a. The camber line ordinate is shifted so that this point coincides with the origin of the local section plane, $\hat{f}(\eta) = f(\eta) - f(\eta_a)$, and the chordwise offset from the stacking axis is defined as $a(\eta) = \eta - \eta_a$.

2.4. Analytical Mid-surface Parametrically Mapped Derivatives

The 3D mid-surface position vector $\mathbf{r}(x,\eta)$ of any point on the blade shell is given by:

$$\mathbf{r}(x,\eta) = (R_h + x)\,\mathbf{e}_X + c(x)\big[a(\eta)\,\mathbf{e}_y(x) + \hat{f}(\eta)\,\mathbf{e}_z(x)\big]. \tag{5}$$

To streamline partial differentiation, two orthogonal section auxiliary vectors $\mathbf{v}(x,\eta)$ and $\mathbf{w}(x,\eta)$ are introduced:

$$\mathbf{v}(x,\eta) = a(\eta)\mathbf{e_y}(x) + \hat{f}(\eta)\mathbf{e_z}(x), \qquad \mathbf{w}(x,\eta) = a(\eta)\mathbf{e_z}(x) - \hat{f}(\eta)\mathbf{e_y}(x). \tag{6}$$

Because the section triad differentiates as $\partial\mathbf{e_y}/\partial x = \kappa\mathbf{e_z}$ and $\partial\mathbf{e_z}/\partial x = -\kappa\mathbf{e_y}$, these auxiliary vectors satisfy $\partial\mathbf{v}/\partial x = \kappa\mathbf{w}$ and $\partial\mathbf{w}/\partial x = -\kappa\mathbf{v}$.

The first covariant mid-surface tangent vectors are evaluated analytically as:

$$\mathbf{r}_{,x} = \frac{\partial\mathbf{r}}{\partial x} = \mathbf{e_X} + c'(x)\mathbf{v} + c(x)\kappa\mathbf{w}, \qquad \mathbf{r}_{,\eta} = \frac{\partial\mathbf{r}}{\partial\eta} = c(x)\big(\mathbf{e_y} + f'(\eta)\mathbf{e_z}\big), \tag{7}$$

where $c'(x) = -c_0\tau_c/L$ is the constant chord taper derivative along the span. Differentiating Eq. (7) yields the second partial surface derivatives:

$$\begin{gathered} \mathbf{r}_{,xx} = 2c'(x)\kappa\mathbf{w} - c(x)\kappa^2\mathbf{v} \\ \mathbf{r}_{,x\eta} = c'(x)\big(\mathbf{e_y} + f'(\eta)\mathbf{e_z}\big) + c(x)\kappa\big(\mathbf{e_z} - f'(\eta)\mathbf{e_y}\big) \\ \mathbf{r}_{,\eta\eta} = c(x)f''(\eta)\mathbf{e_z}. \end{gathered} \tag{8}$$

The unnormalized surface normal vector is $\mathbf{g} = \mathbf{r}_{,x} \times \mathbf{r}_{,\eta}$, the surface Jacobian determinant is $j = \| \mathbf{g} \| > 0$, and the unit normal vector $\mathbf{n}$ is:

$$\mathbf{n} = \frac{\mathbf{r}_{,x} \times \mathbf{r}_{,\eta}}{j}. \tag{9}$$

Differentiating Eq. (9) with respect to surface parametric coordinates $\alpha \in \{x,\eta\}$ and projecting out the normal component yields the spatial derivatives of the unit normal vector:

$$\mathbf{n}_{,\alpha} = \frac{\mathbf{1}}{j}\big[\mathbf{g}_{,\alpha} - \mathbf{n}\big(\mathbf{n}\cdot\mathbf{g}_{,\alpha}\big)\big] \tag{10}$$

where the partial derivatives of the unnormalized surface normal cross-product $\mathbf{g}_{,x}$ and $\mathbf{g}_{,\eta}$ are given by:

$$\mathbf{g}_{,x} = \mathbf{r}_{,xx} \times \mathbf{r}_{,\eta} + \mathbf{r}_{,x} \times \mathbf{r}_{,x\eta}, \qquad \mathbf{g}_{,\eta} = \mathbf{r}_{,x\eta} \times \mathbf{r}_{,\eta} + \mathbf{r}_{,x} \times \mathbf{r}_{,\eta\eta}. \tag{11}$$

2.5. Local Orthonormal Surface Frame and Analytic Frame Derivatives

At every mid-surface point, a right-handed orthonormal triad $\mathbf{Q} = [\mathbf{e}_1\ \mathbf{e}_2\ \mathbf{e}_3]$ is constructed with

$$\mathbf{e}_1 = \frac{\mathbf{r}_{,x}}{\| \mathbf{r}_{,x} \|}, \qquad \mathbf{e}_3 = \mathbf{n}, \qquad \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_1. \tag{12}$$

The spatial partial derivatives of the frame $\mathbf{Q}_{,x}$ and $\mathbf{Q}_{,\eta}$(or $\mathbf{Q}_{,\alpha}$ for $\alpha \in \{x, \eta\}$) are evaluated analytically using the same projection operator applied in Eq. (10) alongside the product rule for the cross product:

$$\mathbf{e}_{1,\alpha} = \frac{1}{\| \mathbf{r}_{,x} \|}(\mathbf{I} - \mathbf{e}_1\mathbf{e}_1^{\mathrm{T}})\mathbf{r}_{,x\alpha}, \qquad \mathbf{e}_{3,\alpha} = \mathbf{n}_{,\alpha}, \qquad \mathbf{e}_{2,\alpha} = \mathbf{n}_{,\alpha} \times \mathbf{e}_1 + \mathbf{n} \times \mathbf{e}_{1,\alpha}, \tag{13}$$

where $\mathbf{I}$ is the 3×3 identity matrix and $\mathbf{e}_1\mathbf{e}_1^{\mathrm{T}}$ denotes the outer product projection tensor.

The availability of $\mathbf{Q}_{,\alpha}$ in exact closed form is essential to the degenerated continuum formulation: the director rotations of the First-Order Shear Deformation Theory (FSDT) kinematics are resolved directly within this local frame, and their spatial derivatives require the spatial derivatives of the frame itself. Consequently, no Lamé parameters (A_1, A_2), Christoffel symbols, or principal curvature tensors (R_1, R_2) appear anywhere in the formulation. The shell geometry is specified entirely by the mid-surface position vector $\mathbf{r}$ and its first and second analytic derivatives $(\mathbf{r}_{,\alpha}, \mathbf{r}_{,\alpha\beta})$.

3. Mathematical Formulation

3.1. Degenerated Continuum Kinematics and FSDT Displacement Field

3.1.1. Degenerated 3D Continuum Geometry and Velocity Field

The 3D spatial position vector $\mathbf{X}(x, \eta, \zeta)$ of any material point within the solid blade volume in the reference configuration is expressed as:

$$\mathbf{X}(x, \eta, \zeta) = \mathbf{r}(x, \eta) + \zeta\, \mathbf{n}(x, \eta), \tag{14}$$

where $\mathbf{r}(x, \eta)$ is the mid-surface position vector, $\mathbf{n}(x, \eta)$ is the unit surface normal, and $\zeta \in [-h/2, h/2]$ is the through-thickness coordinate.

The exact 3D Jacobian matrix $\mathbf{J}$ of the parametric mapping $(x, \eta, \zeta) \mapsto \mathbf{X}$ is evaluated by differentiating Eq. (14) with respect to (x, η, ζ):

$$\mathbf{J} = \left[\, \mathbf{r}_{,x} + \zeta\, \mathbf{n}_{,x} \middle|\, \mathbf{r}_{,\eta} + \zeta\, \mathbf{n}_{,\eta} \middle|\, \mathbf{n} \,\right], \qquad \det \mathbf{J} > 0. \tag{15}$$

The determinant $\det \mathbf{J}$ represents the local volume transformation ratio from the parametric domain to 3D physical space. In the numerical solver, $\det \mathbf{J}$ is monitored at every 3D quadrature point; a non-positive value $(\det \mathbf{J} \leq 0)$indicates that the local shell thickness exceeds the minimum radius of curvature, signaling geometric self-intersection where the degenerated continuum description loses physical validity.

3.1.2. Five-Field FSDT Displacement Field and Spatial Strain Operators

The displacement field follows a five-parameter First-Order Shear Deformation Theory (FSDT) [36,37,51]:

$$\mathbf{u}(x, \eta, \zeta) = \mathbf{u}_0(x, \eta) + \zeta[\phi_1(x, \eta)\, \mathbf{e}_1(x, \eta) + \phi_2(x, \eta)\, \mathbf{e}_2(x, \eta)], \tag{16}$$

where $\mathbf{u}_0 = (u, v, w)^{\mathrm{T}}$ denotes the vector of global Cartesian translations of the mid-surface, and ϕ_1, ϕ_2 are the director rotations resolved in the local tangent frame $(\mathbf{e}_1, \mathbf{e}_2)$. The drilling rotation is omitted, and the director is inextensible $(\varepsilon_{33} = 0)$, which is consistent with the plane-stress material constitutive law adopted below.

The partial derivatives of the displacement field with respect to the parametric coordinates are evaluated analytically as:

$$\mathbf{u}_{,x} = \mathbf{u}_{0,x} + \zeta\big(\phi_{1,x}\mathbf{e}_1 + \phi_1\mathbf{e}_{1,x} + \phi_{2,x}\mathbf{e}_2 + \phi_2\mathbf{e}_{2,x}\big), \qquad \mathbf{u}_{,\zeta} = \phi_1\mathbf{e}_1 + \phi_2\mathbf{e}_2, \tag{17}$$

with the spanwise derivative $\mathbf{u}_{,\eta}$ defined analogously. The frame-derivative terms $\phi_\alpha\mathbf{e}_{\alpha,x}$ and $\phi_\alpha\mathbf{e}_{\alpha,\eta}$ capture the exact kinematic coupling between director rotations and the 3D twist, taper, and camber of the shell surface. These are precisely the terms that classical shell theories approximate through assumed curvature tensors and Lamé parameters.

The global 3D Cartesian displacement gradient tensor $\nabla_X \mathbf{u}$ is evaluated using the inverse 3D Jacobian:

$$\nabla_X \mathbf{u} = \left[\mathbf{u}_{,x}\ \mathbf{u}_{,\eta}\ \mathbf{u}_{,\zeta}\right]\mathbf{J}^{-1}. \tag{18}$$

Transforming this gradient into the local orthonormal material frame $\mathbf{Q} = [\mathbf{e}_1\ \ \mathbf{e}_2\ \ \mathbf{e}_3]$ yields the local displacement gradient matrix $\mathbf{H}$:

$$\mathbf{H} = \mathbf{Q}^{\mathsf{T}}(\nabla_X \mathbf{u})\mathbf{Q}. \tag{19}$$

In the computational implementation, the product $\mathbf{J}^{-1}\mathbf{Q}$ is evaluated via one 3×3 linear system solve per quadrature point, after which the spatial gradient along the local material directions is directly available for constructing both the strain–displacement operator matrix and the initial-stress geometric stiffness matrix.

The 5-component 3D engineering strain vector $\boldsymbol{\varepsilon}$ used with the plane-stress constitutive law is extracted from $\mathbf{H}$ as:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \\ \gamma_{13} \\ \gamma_{23} \end{bmatrix} = \begin{bmatrix} H_{11} \\ H_{22} \\ H_{12} + H_{21} \\ H_{13} + H_{31} \\ H_{23} + H_{32} \end{bmatrix} = \mathbf{B}(x, \eta, \zeta)\mathbf{q}\,, \tag{20}$$

where $\mathbf{q}$ is the global vector of Ritz expansion coefficients, and $\mathbf{B}(x, \eta, \zeta)$ is the spatial strain–displacement operator matrix. The transverse normal strain ε_{33} is eliminated by enforcing the plane-stress condition $\sigma_{33} = 0$.

Because ζ is treated as an independent coordinate in the spatial differentiation of Eq. (16), the spanwise and chordwise variation of thickness enters exclusively through the integration domain limits and the volume determinant $\det \mathbf{J}$, rather than through artificial derivatives in the strain operator, a distinction that is automatic in the degenerated continuum formulation but must be imposed manually in resultant-based shell theories.

3.2. Orthonormal Jacobi–Legendre Rayleigh–Ritz Basis Functions

Each of the five generalized displacement fields $\{u, v, w, \varphi_1, \varphi_2\}$ is expanded in the same tensor-product basis of $n_s \times n_c$ functions:

$$\psi_{ij}(x, \eta) = X_i(x)\, Y_j(\eta), \qquad i = 0, \ldots, n_s - 1, \quad j = 0, \ldots, n_c - 1, \tag{21}$$

yielding a total of $n = 5n_s n_c$ global degrees of freedom.

3.2.1. Spanwise Basis Functions and Exact Clamped Enforcement

Mapping the physical spanwise coordinate $x \in [0, L]$ to the non-dimensional interval $s = 2x/L - 1 \in [-1, 1]$, the spanwise basis functions are defined as:

$$X_i(x) = \sqrt{\frac{2i + 3}{8}}\ (1 + s)\, P_i^{(0,2)}(s), \tag{22}$$

where $P_i^{(\alpha,\beta)}(s)$ is the Jacobi polynomial [65,66]. The boundary factor $(1 + s)$ vanishes identically at the root $x = 0$ (where $s = -1$), ensuring that every field (translations and director rotations alike) satisfies the fully clamped kinematic boundary condition identically, without requiring penalty functions or artificial spring constraint equations.

The Jacobi weight function $(1 + s)^\beta$ with $\beta = 2$ naturally absorbs the square of this boundary factor, yielding exact orthonormality:

$$\int_{-1}^{1} X_i(s)\, X_k(s)\, ds = \delta_{ik}, \tag{23}$$

by virtue of the Jacobi polynomial orthogonality relation:

$$\int_{-1}^{1} (1 + s)^2 P_i^{(0,2)}(s) P_k^{(0,2)}(s)\, ds = \frac{8}{2i + 3}\delta_{ik}.$$

The spatial derivative with respect to s is evaluated analytically using:

$$\frac{d}{ds}P_i^{(0,2)}(s) = \frac{1}{2}(i+3)P_{i-1}^{(1,3)}(s).$$

The spanwise basis thus forms a set of L_2-orthonormal admissible functions in the sense of Bhat [46] and Liew and Lam [47]. Because these functions are obtained in exact closed form rather than via Gram–Schmidt numerical orthogonalization, they avoid the loss of orthogonality and numerical ill-conditioning that accumulate in generated recurrence sequences [50].

3.2.2. Chordwise Basis Functions and Free Boundary Conditions

Mapping the non-dimensional chordwise coordinate $\eta \in [0,1]$ to $t = 2\eta - 1 \in [-1,1]$, the normalized chordwise basis functions are defined as:

$$Y_j(\eta) = \sqrt{\frac{2j+1}{2}}P_j(t), \qquad j = 0, \dots, n_c - 1, \tag{24}$$

where $P_j(t)$ denotes the standard Legendre polynomial of degree j(equivalent to the Jacobi polynomial $P_j^{(0,0)}(t)$). The chordwise functions satisfy exact L_2-orthonormality:

$$\int_0^1 Y_j(\eta)Y_l(\eta)\,d\eta = \delta_{jl}.$$

Spatial derivatives with respect to t are evaluated analytically using the Jacobi derivative identity:

$$\frac{d}{dt}P_j(t) = \frac{1}{2}(j+1)P_{j-1}^{(1,1)}(t).$$

The leading ($\eta = 0$) and trailing ($\eta = 1$) edges are physically free; the corresponding natural boundary conditions are satisfied variationally through the Rayleigh–Ritz energy minimization.

The combined L_2-orthonormality of both the spanwise $X_i(x)$ and chordwise $Y_j(\eta)$ basis families produces a global mass matrix that, for a flat plate of uniform thickness and orthogonal parametric lines, would be strictly diagonal. For the actual twisted, tapered airfoil shell, the mass matrix remains strongly diagonally dominant. This structural property underpins the well-conditioned mass-normalization procedure detailed in Section 3.6 and represents a decisive advantage over standard monomial or trigonometric Ritz bases, whose Gram matrices become severely ill-conditioned or numerically singular at moderate polynomial orders.

Continuum Displacement Operators: For the translational fields ($k = 1,2,3$), the shape operator is $\mathbf{U}^{(k)} = \mathbf{e}_k \otimes \boldsymbol{\psi}$ with constant global Cartesian directions $\mathbf{e}_k$. For the rotational fields ($\alpha = 1,2$), it is $\mathbf{U}^{(3+\alpha)} = \zeta\mathbf{e}_\alpha(x,\eta) \otimes \boldsymbol{\psi}$, whose parametric derivatives require the local frame derivatives $\mathbf{e}_{\alpha,x}$ and $\mathbf{e}_{\alpha,\eta}$ established in Section 2.5. The partial derivatives in Eq. (14) are evaluated at fixed η, consistent with the surface parametrization.

3.3. Constitutive Relations and Rotating Energy Functionals

3.3.1. Plane-Stress Elasticity and Transverse Shear Rigidity

For an isotropic, homogeneous material with Young's modulus E, Poisson's ratio ν, and shear modulus $G = E/[2(1+\nu)]$, the plane-stress constitutive matrix $\mathbf{C}$, in the local frame is:

$$\mathbf{C} = \begin{bmatrix} \frac{E}{1-\nu^2} & \frac{\nu E}{1-\nu^2} & 0 & 0 & 0 \\ \frac{\nu E}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 & 0 & 0 \\ 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & k_s G & 0 \\ 0 & 0 & 0 & 0 & k_s G \end{bmatrix}, \tag{25}$$

with the shear correction factor $k_s = \pi^2/12$ [37, 61]. The transverse shear correction compensates for the constant through-thickness distribution of γ_{13}, γ_{23} implied by Eq. (16).

3.3.2. Rotating Energy Functionals and System Matrices

Let $\mathbf{S}$ denote the skew-symmetric spin matrix corresponding to the angular velocity vector $\boldsymbol{\Omega} = \Omega\mathbf{e}_Y$, satisfying $\mathbf{Sa} = \boldsymbol{\Omega} \times \mathbf{a}$ for any vector $\mathbf{a}$:

$$\mathbf{S} = \begin{bmatrix} 0 & 0 & \Omega \\ 0 & 0 & 0 \\ -\Omega & 0 & 0 \end{bmatrix}, \qquad \mathbf{S}^{\mathrm{T}}\mathbf{S} = -\mathbf{S}^2. \tag{26}$$

The absolute acceleration $\mathbf{a}$ of a material point observed within the rotating reference frame is expressed as:

$$\mathbf{a} = \ddot{\mathbf{u}} + 2\mathbf{S}\dot{\mathbf{u}} + \mathbf{SS}(\mathbf{X} + \mathbf{u}) \tag{27}$$

comprising the relative acceleration $\ddot{\mathbf{u}}$, the Coriolis acceleration $2\mathbf{S}\dot{\mathbf{u}}$, and the centripetal acceleration $\mathbf{S}^2(\mathbf{X} + \mathbf{u})$.

Expressing the continuous displacement field in discretized form as $\mathbf{u}(x, \eta, \zeta, t) = \mathbf{U}(x, \eta, \zeta)\mathbf{q}(t)$, where $\mathbf{U}$ is the $3 \times n$ global displacement operator constructed from the Ritz basis polynomials and local director frame, application of D'Alembert's principle of virtual work:

$$\int_V \rho\ \delta\mathbf{u}^{\mathrm{T}}\mathbf{a}\, dV + \int_V \delta\,\boldsymbol{\varepsilon}^{\mathrm{T}}\boldsymbol{\sigma}\, dV = 0, \tag{28}$$

yields the semi-discrete non-linear governing equations of motion:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{G}\dot{\mathbf{q}} + (\mathbf{K}_{\mathrm{el}} + \mathbf{K}_\sigma - \mathbf{K}_\Omega)\mathbf{q} = \mathbf{f}_c, \tag{29}$$

with the elastic stiffness matrix $\mathbf{K}_{\mathrm{el}}$, mass matrix $\mathbf{M}$, gyroscopic matrix $\mathbf{G}$, spin-softening matrix $\mathbf{K}_\Omega$, and centrifugal body-force load vector $\mathbf{f}_c$ defined by the 3D volume integrals:

$$\begin{aligned} \mathbf{K}_{\mathrm{el}} &= \int_V \mathbf{B}^{\mathrm{T}}\,\mathbf{C}\,\mathbf{B}\, dV, & \mathbf{M} &= \int_V \rho\ \mathbf{U}^{\mathrm{T}}\mathbf{U}\, dV, \\ \mathbf{G} &= 2\int_V \rho\ \mathbf{U}^{\mathrm{T}}\mathbf{S}\,\mathbf{U}\, dV, & \mathbf{K}_\Omega &= \int_V \rho\ \mathbf{U}^{\mathrm{T}}\mathbf{S}^{\mathrm{T}}\mathbf{S}\,\mathbf{U}\, dV, \\ \mathbf{f}_c &= \int_V \rho\ \mathbf{U}^{\mathrm{T}}\mathbf{S}^{\mathrm{T}}\mathbf{S}\,\mathbf{X}\, dV, \end{aligned} \tag{30}$$

where the differential volume element dV is mapped to the normalized parametric domain $\hat{\zeta} \in [-1, 1]$ as:

$$dV = \det\mathbf{J}\ \frac{h(x, \eta)}{2}\, d\hat{\zeta}\, d\eta\, dx\,. \tag{31}$$

The initial-stress geometric stiffness matrix $\mathbf{K}_\sigma$ depends on the static centrifugal stress field and is derived in detail in Section 3.5. The mass matrix $\mathbf{M}$, elastic stiffness matrix $\mathbf{K}_{\mathrm{el}}$, and spin-softening matrix $\mathbf{K}_\Omega$ are inherently symmetric ($\mathbf{M}^{\mathrm{T}} = \mathbf{M}$, $\mathbf{K}_{\mathrm{el}}^{\mathrm{T}} = \mathbf{K}_{\mathrm{el}}$, $\mathbf{K}_\Omega^{\mathrm{T}} = \mathbf{K}_\Omega$), whereas the gyroscopic matrix $\mathbf{G}$ is skew-symmetric ($\mathbf{G}^{\mathrm{T}} = -\mathbf{G}$); these exact structural symmetries are enforced post-assembly to eliminate numerical round-off asymmetries. The negative sign of $\mathbf{K}_\Omega$ reflects the physical phenomenon of spin softening: the centripetal term $\mathbf{S}^2\mathbf{u} = -\mathbf{S}^{\mathrm{T}}\mathbf{S}\mathbf{u}$ acts to reduce the effective dynamic restoring stiffness for structural displacements perpendicular to the spin axis. Furthermore, because the displacement operator $\mathbf{U}$ naturally incorporates the ζ-weighted director rotation components from FSDT kinematics, both the translational and rotary inertia of the shell cross-section are captured consistently across $\mathbf{M}$, $\mathbf{G}$, and $\mathbf{K}_\Omega$, and the centrifugal body load vector $\mathbf{f}_c$ inherently includes the steady overturning moment generated by the through-thickness offset of the body force.

3.4. Singularity-Aware 3D Volume Quadrature

The volume integrands defining the mass, stiffness, and load matrices in Eqs. (30) and (31) (as well as the initial-stress geometric stiffness in Eq. (34)) are smooth functions along the spanwise coordinate x, so $n_q^{(x)}$ Gauss–Legendre quadrature points distributed over $[0, L]$ suffice for exact numerical integration. In the chordwise direction η, however, two distinct geometric features severely degrade the convergence rate of standard Gauss quadrature:

i. **Leading-edge thickness singularity:** The square-root behavior $\sqrt{\eta}$ in the NACA thickness law [Eq. (2)] enters the volume element dV through $h(x, \eta)/2$ [Eq. (31)] and the stiffness/mass operators through hand

h^3-type thickness moments, causing the chordwise thickness derivative to blow up at the leading edge ($\partial h_0/\partial \eta \to \infty$ as $\eta \to 0$).

ii. **Camber-curvature jump:** The second derivative of the camber line $f''(\eta)$ undergoes a finite jump discontinuity across the maximum-camber location $\eta = p$ [Eq. (1)]. This discontinuity propagates directly into $\mathbf{r}_{,\eta\eta}$ [Eq. (8)], the surface normal derivatives $\mathbf{n}_{,\eta}$ [Eq. (10)], the local frame derivatives $\mathbf{Q}_{,\eta}$ [Eq. (13)], and the 3D volume Jacobian determinant $\det \mathbf{J}$ [Eq. (15)].

To restore exponential (spectral) quadrature convergence, a two-part coordinate transformation and domain subdivision strategy is implemented:

Leading-Edge Regularization Mapping ($\eta = t^2$): The coordinate substitution $\eta = t^2$ (with $d\eta = 2t\, dt$) maps $\sqrt{\eta} \to t$, transforming the non-smooth square-root singularity into a smooth linear polynomial in $t \in [0, 1]$. The mapping Jacobian $2t$ is absorbed directly into the chordwise quadrature weights $\left(w_\eta = 2t\, w_t\right)$. Standard Gauss–Legendre quadrature in tthen converges exponentially.

Camber-Curvature Domain Subdivision: Because $\eta = t^2$, the maximum-camber location $\eta = p$ corresponds to $t = \sqrt{p}$. The integration interval $t \in [0, 1]$ is divided at $t = \sqrt{p}$ into two sub-intervals, $\left[0, \sqrt{p}\,\right]$ and $\left[\sqrt{p}, 1\right]$, so that the camber line within each sub-interval is polynomially smooth. On each sub-interval, $n_q^{(\eta)}$Gauss–Legendre points are placed independently. For symmetric profiles ($m = 0$), no split is required and a single integration domain $t \in [0, 1]$ is used.

Through-Thickness Layered Quadrature: Through the thickness, $n_q^{(\zeta)}$ Gauss–Legendre points are placed on $\zeta \in [-1, 1]$, where physical thickness coordinates are mapped via $z = \zeta\, h(x, \eta)/2$. The integrands in ζ are at most quadratic polynomials multiplied by the rational function $\det \mathbf{J}\,(\zeta)$ [Eq. (15)]. Setting $n_q^{(\zeta)} \geq 2$ (typically $n_q^{(\zeta)} = 3$) guarantees that both translational and rotary inertia terms are captured with high precision.

The total number of 3D volume quadrature points is $n_q^{(x)} \times n_q^{(\eta)} \times n_{sub} \times n_q^{(\zeta)}$ (where $n_{sub} = 2$ for cambered profiles and 1 for symmetric profiles). To prevent numerical rank deficiency in the discretized global matrices, the quadrature orders satisfy $n_q^{(x)} \geq n_s + 1$ and $n_q^{(\eta)} \geq n_c + 1$.

3.5. Prestress-Consistent Geometric Stiffness and Stress Recovery

Centrifugal stiffening arises from the interaction of the steady stress field with the second-order (rotation) part of the strain. Rather than adopting a beam-type axial-force approximation or a membrane-resultant approximation, the present formulation evaluates the complete three-dimensional stress state produced by the centrifugal load and integrates it against the full displacement gradient.

Static equilibrium. The steady deformation $\mathbf{q}_0$ on the reference geometry is obtained from:

$$\mathbf{K}_{el}\, \mathbf{q}_0 = \mathbf{f}_c, \tag{32}$$

solved in mass-normalized coordinates after verification that the balanced stiffness admits a Cholesky factorization; failure indicates that the reference configuration is not a stable static state and that a nonlinear equilibrium path would be required. The relative residual $(\|\mathbf{K}_{el}\, \mathbf{q}_0 - \mathbf{f}_c\|)/\|\mathbf{f}_c\|$ is recorded as a solution diagnostic.

Stress recovery. At every volume quadrature point, the 5-component strain vector $\boldsymbol{\varepsilon}_0 = \mathbf{B}\mathbf{q}_0$ and Cauchy stress vector $\boldsymbol{\sigma}_0 = \mathbf{C}\boldsymbol{\varepsilon}_0$ are evaluated and assembled into the symmetric local stress tensor $\boldsymbol{\Sigma}_0$:

$$\boldsymbol{\Sigma}_0 = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & 0 \end{bmatrix}. \tag{33}$$

Initial-stress stiffness. Linearizing the Green–Lagrange strain about the prestressed state and retaining the quadratic term yields the classical initial-stress operator [41,45,62]:

$$\delta W_\sigma = \int_V \sigma_{ab}^0\ \delta u_{k,a}\, u_{k,b}\, dV \quad \Rightarrow \quad \mathbf{K}_\sigma = \int_V \sum_{k=1}^{3} (\nabla_\ell U_k)^T\, \boldsymbol{\Sigma}_0\, (\nabla_\ell U_k)\, dV, \tag{34}$$

where $\nabla_l U_k$ is the row of the operator giving the derivative of the k-th Cartesian displacement component along the three local directions. All nine gradient components and all five independent stress components participate; no resultant or beam approximation is introduced. This is the sense in which the formulation is prestress-consistent: the same kinematic operator, the same quadrature, and the same three-dimensional Jacobian are used for the static solution, the stress recovery, and the geometric stiffness.

Validity of the linear reference-state approximation. Equation (32) is a linear solve on the undeformed geometry[4]. Its adequacy is checked a posteriori by three quantities reported for every analysis: the maximum linear strain $\max|\boldsymbol{\varepsilon}_0|$(threshold set at 1%), the maximum static mid-surface displacement normalized by span (threshold set at 2%), and the maximum director slope $\sqrt{\phi_1^2 + \phi_2^2}$ (threshold set at 0.05 rad). Exceedance of any threshold indicates that a geometrically non-linear equilibrium and tangent operator would be required.

3.6. Mass Normalization and Energy-Scaled Gyroscopic Eigensolver

Setting $\mathbf{f}_c$ aside (it determines the static prestress deformation $\mathbf{q}_0$ only) and substituting $\mathbf{q}(t) = \boldsymbol{\phi} e^{\lambda t}$ into the homogeneous form of Eq. (29) yields the quadratic eigenvalue problem:

$$(\lambda^2 \mathbf{M} + \lambda\, \mathbf{G} + \mathbf{K})\boldsymbol{\phi} = \mathbf{0}, \qquad \mathbf{K} = \mathbf{K}_{el} + \mathbf{K}_\sigma - \mathbf{K}_\Omega. \tag{35}$$

where $\lambda \in \mathbb{C}$ is the complex eigenvalue, $\boldsymbol{\phi} \in \mathbb{C}^n$ is the corresponding eigenvector (mode shape), and $\mathbf{K}$ represents the combined effective stiffness matrix.

The natural cyclic frequency of a mode is $f = \mathrm{Im}(\lambda)/2\pi$ (in Hz) and its growth/decay rate is given by $\mathrm{Re}(\lambda)$. For $K \succ 0$ (strictly positive-definite total stiffness matrix), the governing system behaves as a conservative gyroscopic system whose eigenvalues are purely imaginary ($\mathrm{Re}(\lambda) = 0$) and whose motion is dynamically stable [55, 56, 63, 64]. Loss of positive-definiteness of K(which spin softening K_Ω can produce at sufficiently high rotational speeds) admits eigenvalues with a positive real part ($\mathrm{Re}(\lambda) > 0$), corresponding to divergence-type dynamic instability. Detecting and quantifying this stability boundary is a primary objective of the present solver.

To preserve numerical precision across the disparate scales of translational and ζ-weighted rotational degrees of freedom, the mass matrix is diagonally equilibrated, $\widetilde{\mathbf{M}} = \mathbf{DMD}$ with $\mathbf{D} = \mathrm{diag}\left(M_{ii}^{-1/2}\right)$, and factorized via Cholesky decomposition, $\widetilde{\mathbf{M}} = \mathbf{L}\mathbf{L}^T$. The congruence transformation $\mathbf{q} = \mathbf{T}\mathbf{y}$ with $\mathbf{T} = \mathbf{D}\mathbf{L}^{-T}$ reduces the quadratic gyroscopic eigenvalue problem to standard mass-orthonormal form:

$$\left(\lambda^2 \mathbf{I} + \lambda \widetilde{\mathbf{G}} + \widetilde{\mathbf{K}}\right)\mathbf{y} = 0,$$

where $\widetilde{\mathbf{K}} = \mathbf{T}^T\mathbf{K}\mathbf{T}$ is symmetric and $\widetilde{\mathbf{G}} = \mathbf{T}^T\mathbf{G}\mathbf{T}$ is real skew-symmetric.

When the effective stiffness is positive-definite ($\lambda_{\min}\left(\widetilde{\mathbf{K}}\right) > 0$), the spectral decomposition $\widetilde{\mathbf{K}} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^T$ allows scaling the state vector by the square-root stiffness, $\mathbf{z} = \left\{\boldsymbol{\Lambda}^{1/2}; \mathbf{V}^T\mathbf{y} \;\; \mathbf{V}^T\dot{\mathbf{y}}\right\}$. The first-order state equation becomes:

$$\dot{\mathbf{z}} = \mathbf{A}\mathbf{z},\ \mathbf{A} = \begin{bmatrix} \mathbf{0} & \boldsymbol{\Lambda}^{1/2} \\ -\boldsymbol{\Lambda}^{1/2} & -\mathbf{V}^T\widetilde{\mathbf{G}}\mathbf{V} \end{bmatrix},$$

where Ais strictly real skew-symmetric ($\mathbf{A}^T = -\mathbf{A}$). Consequently, i$\mathbf{A}$ is Hermitian, ensuring that a standard LAPACK real symmetric eigensolver (scipy.linalg.eigh) returns purely imaginary complex eigenvalues ($\lambda = -i\mu$) with zero spurious real parts. Physical modal vectors are recovered via $\boldsymbol{\phi} = \mathbf{T}\mathbf{V}\boldsymbol{\Lambda}^{-1/2}\mathbf{z}_{(1)}$.

If spin-softening causes stiffness loss of positive-definiteness ($\lambda_{\min}\left(\widetilde{\mathbf{K}}\right) \le 0$), the solver transitions to a time-scaled $2n \times 2n$ companion state formulation solved via a general non-symmetric LAPACK eigensolver (scipy.linalg.eig), where positive real parts ($\mathrm{Re}(\lambda) > 10^{-7}$) identify growing divergence-type instabilities. All computed eigenpairs $(\lambda, \boldsymbol{\phi})$ are normalized to unit modal mass and verified by evaluating relative residuals ($r \le 10^{-6}$) on the unconstrained global matrices.

4. Computational and Numerical Implementation

4.1. One-Pass Matrix Assembly and Volume Check

All linear system operators defining the unconstrained dynamic system are assembled in a single vectorized pass over the 3D volume quadrature points. To strictly bound peak memory usage, quadrature points are evaluated layer-by-layer along the through-thickness coordinate $\zeta \in [-1, 1]$. At each layer, physical coordinates $z = \zeta h(x, \eta)/2$ are assigned across surface quadrature points (x, η). The displacement shape operator $\mathbf{U}$, local gradient $\nabla_1 \mathbf{U}$, strain-displacement matrix $\mathbf{B}$, Jacobian determinant $\det \mathbf{J}$ [Eq. (15)], and material position $\mathbf{X}$ [Eq. (14)] are evaluated in vectorized form. The elemental contributions

$$\mathbf{B}^\top \mathbf{C} \mathbf{B}, \quad \rho\, \mathbf{U}^\top \mathbf{U}, \quad 2\rho\, \mathbf{U}^\top \mathbf{S} \mathbf{U}, \quad \rho\, (\mathbf{S}\mathbf{U})^\top (\mathbf{S}\mathbf{U}), \quad \rho\, \mathbf{U}^\top \mathbf{S}^\top \mathbf{S} \mathbf{X}$$

are accumulated into global system matrices using effective weights $w = w_x w_\eta w_\zeta\, h(x, \eta)/2 \det \mathbf{J}$, yielding $\mathbf{K}_{el}$, $\mathbf{M}$, $\mathbf{G}$, $\mathbf{K}_\Omega$, and $\mathbf{f}_c$ [Eq. (30)]. Simultaneously, physical blade volume $V = \int dV$ and mass $m = \rho V$ are accumulated and verified against the CAD model to confirm mapping accuracy and Jacobian positivity ($\det \mathbf{J} > 0$). Following the static solve $\mathbf{q}_0$ [Eq. (32)], a second vectorized pass evaluates Cauchy stress $\mathbf{\Sigma}_0$ [Eq. (33)] and assembles geometric stiffness $\mathbf{K}_\sigma$ [Eq. (34)].

4.2. Software Stack and Reproducibility

The formulation is implemented in Python using NumPy [70] and SciPy [71] (utilizing LAPACK-backed dense linear algebra [68] for Jacobi and Legendre polynomial evaluation) with Matplotlib [72] for post-processing. For each numerical simulation, the code exports the modal frequency table with growth rates and relative residuals, the complete complex eigen spectrum, the assembled global system operators, the static coefficient vector, and a JSON record summarizing all configuration parameters and numerical diagnostics. Root and tip section point clouds are exported in a CAD-compatible tabular format to permit the construction of a geometrically identical 3D solid model in commercial software for finite element validation.

4.3. Baseline Reference Blade Configuration and CAD Export

To evaluate the convergence, numerical accuracy, and dynamic characteristics of the proposed semi-analytical framework, a baseline cantilevered compressor blade with a pre-twisted, tapered, cambered NACA 6412 profile is defined. The baseline geometry, material properties, rotational operating parameters, and numerical discretization orders used across the benchmark studies in Section 5 are summarized in Table 1 below.

Table 1. Reference blade physical parameters, material properties, and Ritz discretization settings.

Category	Quantity	Symbol	Value
Blade Geometry	Span length	L	0.3 m
	Root chord	c_0	0.2 m
	Cross-section profile	—	NACA 6412
	Chord taper fraction	τ_c	0.10
	Thickness taper fraction	τ_h	0.50
	Root stagger angle	θ_0	$-20°$
	Total tip pre-twist	θ_{tip}	$-15°$
	Pre-twist axis chord fraction	η_a	0.25(quarter-chord)
	Rotor hub radius	R_h	0.2 m
Operational State	Rotor speed	N (Ω)	4000 rpm
Material Properties	Young's modulus	E	200 GPa
	Poisson's ratio	ν	0.30
	Density	ρ	7850 kg m^{-3}
	Shear correction factor	κ	$\pi^2/12 \approx 0.8225$
Ritz Discretization	Ritz polynomial order	$n_s \times n_c$	6×6 (180 DOFs)
	Volume quadrature points	$n_q^{(x)}, n_q^{(\eta)}, n_q^{(\zeta)}$	$10 \times (10 \times 2) \times 3 = 600$points

5. Results and Discussion

5.1. Geometric Verification and Volume Integration Accuracy

Prior to dynamic and modal analyses, the analytical geometric mapping derived in Section 2 was verified against the 3D CAD solid model constructed in ANSYS. To ensure complete geometric consistency between the semi-analytical solver and the finite element baseline, 12 spanwise cross-sectional station coordinates evaluated in the Python solver were exported to .txt file and imported into ANSYS to loft the 3D CAD solid model (Figure 4). Figure 2 illustrates the cross-sectional profiles generated analytically at the blade root ($x = 0$) and tip ($x = L$). The plot displays the upper (suction) and lower (pressure) skin boundaries, the camber line, the straight chord line, and the pre-twist stacking axis located at the quarter-chord point ($\eta_a = 0.25$). The tip section demonstrates the combined spanwise tapering and pre-twist functions: a 10% chord reduction ($\tau_c = 0.10$), a 50% thickness taper ($\tau_h = 0.50$), and a relative tip pre-twist of $-15°$($\theta_{tip} = -15°$) relative to the $-20°$root stagger angle ($\theta_0 = -20°$).

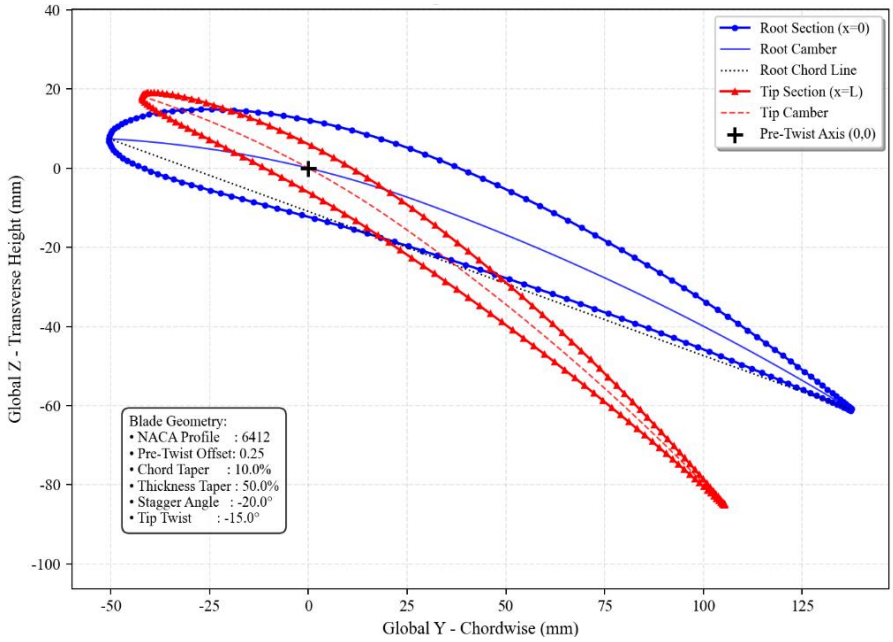


Figure 2. Root and tip section geometry: Overlay of the analytically generated cross-sections at the blade root ($x = 0$, solid curve) and tip ($x = L$, dashed curve) in the local section plane (Y–Z), showing upper and lower skin envelopes, camber lines, straight chord lines, and the quarter-chord stacking axis at $(0, 0)$. The tip profile illustrates the 10% chord reduction, 50% thickness taper, and $-15°$relative pre-twist.

The continuous 3D surface mapping generated by the analytical formulation is compared against the reconstructed solid CAD model in Figure 3 and Figure 4. Figure 3 shows the analytical 3D representation of the blade, illustrating

the smooth, continuous variation of pre-twist, taper, and camber along the span. Figure 4 displays the corresponding 3D solid CAD model created in ANSYS by lofting through the exported station point clouds.

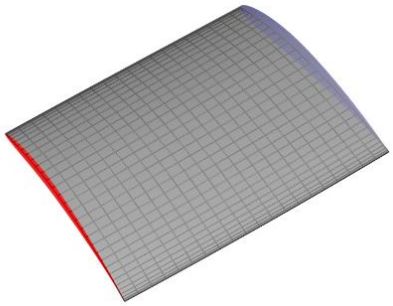


Figure 3. Analytical 3D blade representation

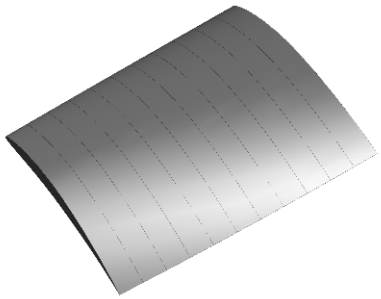


Figure 4. ANSYS CAD model reconstruction

To verify 3D volume integration and the singularity-aware quadrature scheme (Section 3.4), the total integrated blade volume ($V = \int_V \det J_r \, dV$) and mass ($m = \rho V$) were compared against the ANSYS CAD solid model report (Table 2). The Rayleigh–Ritz volume matches the CAD model to within 0.221% relative difference.

Table 2. Comparison of integrated blade volume and mass between the present Rayleigh–Ritz formulation and the 3D ANSYS CAD solid model for the reference NACA 6412 blade.

Parameter	ANSYS CAD Model	Present Rayleigh–Ritz	Relative Difference (%)
Mass (kg)	5.36798535	5.35611030	-0.2212%
Volume (m^3)	0.00068382	0.00068231	-0.2208%

As listed in Table 2, the Rayleigh–Ritz volume matches the ANSYS CAD solid model to within 0.2208% relative difference, and the integrated mass matches to within 0.2212%. This close agreement confirms the accuracy of the exact 3D volume metric tensor and First-Order Shear Deformation Theory (FSDT) integration over complex pre-twisted, tapered geometries.

5.2. Convergence Characteristics of Orthonormal Ritz Polynomials

To ensure numerical stability and determine the optimal expansion order for spectral accuracy, a systematic convergence study was performed by varying the Ritz polynomial order $n_s = n_c$ from 4×4 to 7×7 (corresponding to system DOFs $N = 80$ to $N = 245$). As shown in Table 3, the natural frequencies converge rapidly and monotonically toward the 3D ANSYS finite element baseline.

Table 3. Convergence of natural frequencies (Hz) at N = 4,000 rpm for increasing Ritz order ($n_s \times n_c$) compared against ANSYS FEA.

Ritz Order	Model / Error	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7
4×4	Present (RR)	248.87	724.04	1005.19	1623.39	1734.83	2464.40	2906.82
	ANSYS FEA	245.85	706.85	970.04	1606.69	1659.01	2098.47	2419.68
	Error (%)	1.23%	2.43%	3.62%	1.04%	4.57%	17.44%	20.13%
5×5	Present (RR)	246.86	712.70	972.58	1609.34	1664.66	2226.00	2446.49
	ANSYS FEA	245.85	706.85	970.04	1606.69	1659.01	2098.47	2419.68
	Error (%)	0.41%	0.83%	0.26%	0.17%	0.34%	6.08%	1.11%
6×6	Present (RR)	246.01	708.17	969.46	1606.47	1657.86	2116.92	2413.46
	ANSYS FEA	245.85	706.85	970.04	1606.69	1659.01	2098.47	2419.68
	Error (%)	0.07%	0.19%	-0.06%	-0.01%	-0.07%	0.88%	-0.26%
7×7	Present (RR)	245.66	706.49	968.41	1605.21	1655.06	2103.02	2405.94
	ANSYS FEA	245.85	706.85	970.04	1606.69	1659.01	2098.47	2419.68
	Error (%)	-0.08%	-0.05%	-0.17%	-0.09%	-0.24%	0.22%	-0.57%

At the lowest expansion order of 4×4 (80 DOFs), relative errors with respect to ANSYS range from 1.04% to 20.13% for the lowest seven modes. Increasing the order to 5×5(125 DOFs) markedly improves accuracy, reducing all relative errors below 6.08%. At 6×6(180 DOFs), the present Rayleigh–Ritz formulation achieves essentially complete agreement with ANSYS, with a maximum error of 0.07% for Mode 1 and errors below 0.88% for all primary modes. Further increasing the order to 7×7(245 DOFs) produces negligible frequency changes. Accordingly, an expansion order of $n_s = n_c = 6$($N = 180$ DOFs) offers an optimal balance between spectral accuracy and computational efficiency and is therefore adopted for all subsequent rotating benchmark studies.

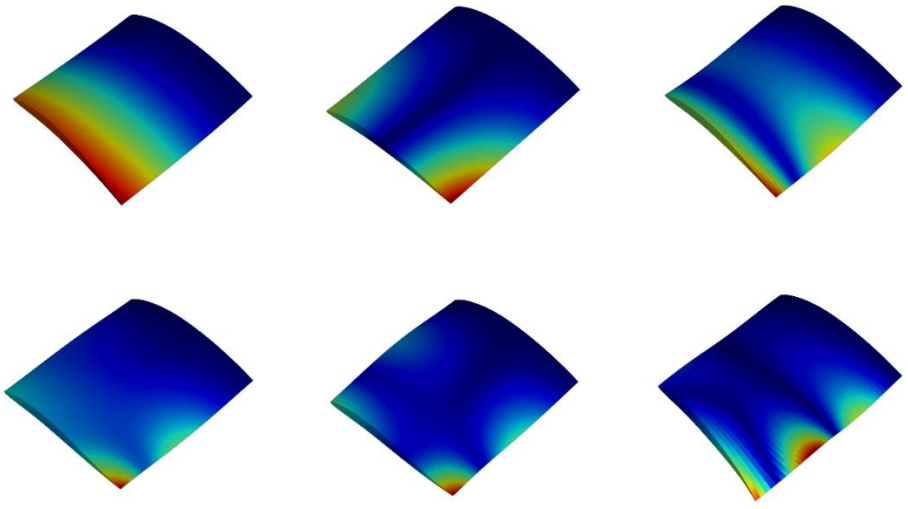
5.3. Non-Rotating Modal Validation against 3D Solid FEA (0 RPM)

Table 4. Non-rotating frequency comparison with ANSYS.

Modes	ANSYS	RR	Error
1	231.0974	231.301	0.09%
2	701.4673	702.82	0.19%
3	950.0262	949.296	-0.08%
4	1601.545	1601.127	-0.03%
5	1647.969	1646.059	-0.12%
6	2075.323	2094.392	0.92%
7	2415.01	2408.762	-0.26%

(RPM: 0.01 rpm, used here)

Caption: Lowest seven natural frequencies (Hz) of the reference blade at $N = 0$: present method (150 DOF) versus ANSYS Mechanical using (a) quadratic shell elements (SHELL281) on the mid-surface with the exact thickness distribution applied as a field, and (b) quadratic solid elements (SOLID186) on the CAD solid. State the element counts and degrees of freedom of each FE model, the mesh-convergence tolerance achieved, and the percentage difference of the present method from each. Discuss the residual differences in terms of the plane-stress assumption, the shear correction factor, and the finite trailing-edge face modelled in the solid.



(From code-RR)

Figure 5 — Non-rotating mode shapes: present versus ANSYS. Caption: Side-by-side deformed skins for the lowest six modes, colored by displacement magnitude on a common scale. Identify each mode by its dominant character (first flap, first edgewise, first torsion, second flap, chordwise bending, coupled flap–torsion) and comment on the flap–torsion coupling induced by pre-twist and the stacking-axis offset from the section centroid. Report the Modal Assurance Criterion between corresponding present and FE mode shapes evaluated on a common grid of mid-surface points.

5.4. Rotating Modal Validation and Prestress Analysis (4,000 RPM and 20,000 RPM)

[PLACEHOLDER: Table 5 — Static prestress diagnostics.] Caption: For N ∈ {1000, ...,8000} rpm: static solve relative residual, maximum linear strain, maximum mid-surface displacement normalised by L, maximum director slope, and the resulting maximum von Mises stress at the root. Compare the root radial stress and tip displacement with the ANSYS static prestress solution (rotational inertia load). Mark the speed, if any, at which a validity threshold of is exceeded.

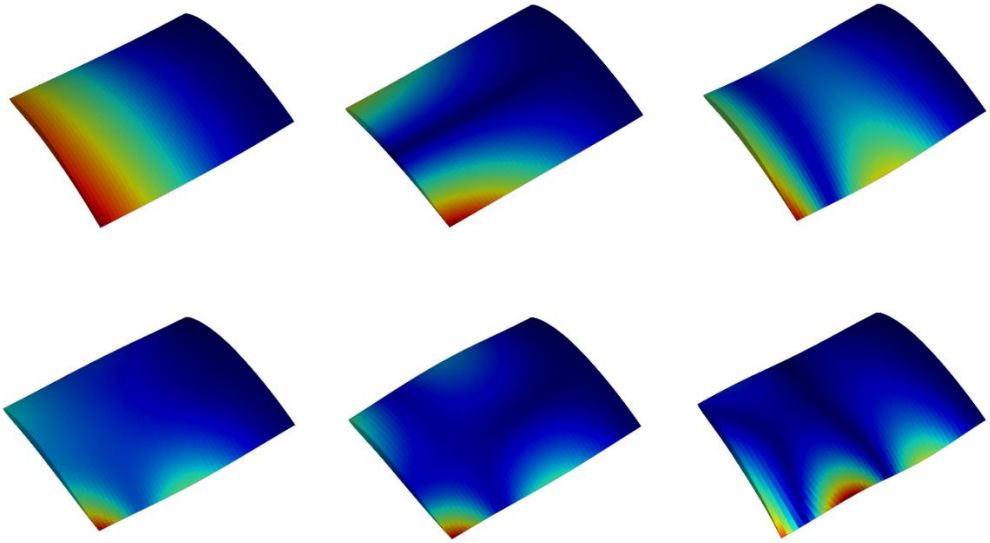
[PLACEHOLDER: Figure 6 — Centrifugal stress field.] Caption: Contours of the mid-surface radial stress σ_{11} and shear stress σ_{12} recovered from $\mathbf{q}_0$ at 4000 rpm, showing the concentration toward the root and the chordwise redistribution induced by pre-twist and stagger. Compare with the corresponding ANSYS contour.

Table 6 — Rotating frequency comparison with ANSYS: Lowest seven frequencies at 4000 rpm: present method (with $\mathbf{K}_\sigma$, $\mathbf{K}_\Omega$, and $\mathbf{G}$) versus ANSYS prestressed modal analysis in the rotating frame with Coriolis effects activated and spin softening included. Provide percentage differences and identify any modal reordering with respect to the non-rotating case.

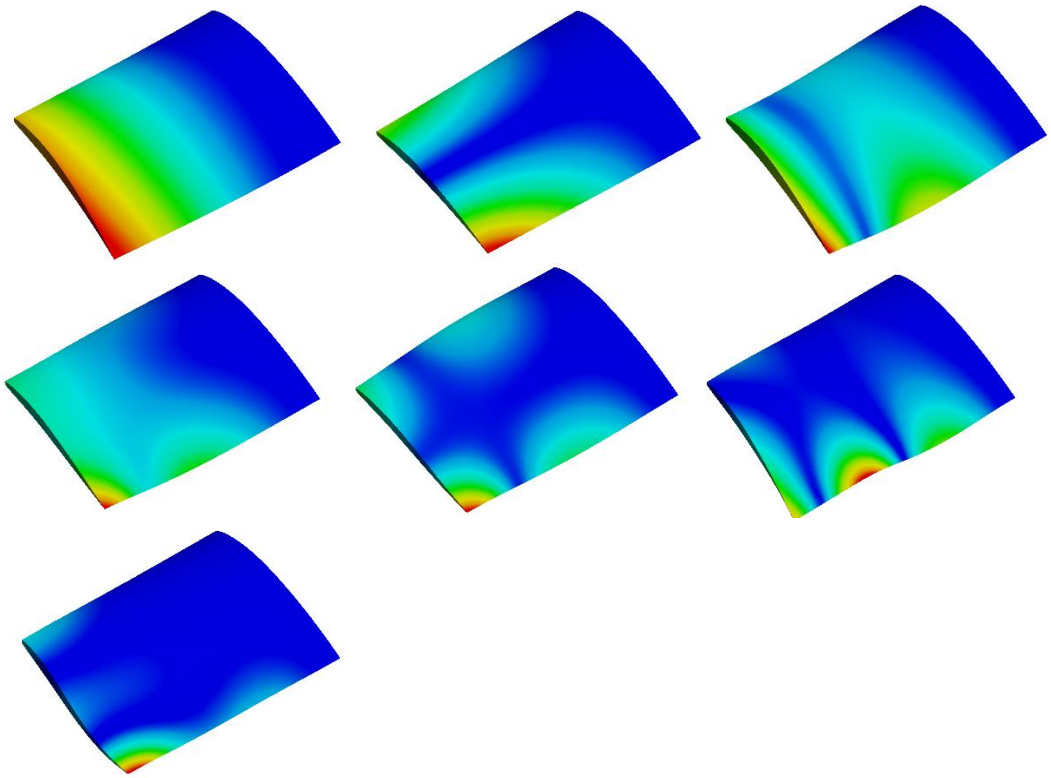
Modes	ANSYS	Present (RR)	Error
1	245.85	246.012	0.07%
2	706.85	708.174	0.19%
3	970.04	969.46	-0.06%
4	1606.69	1606.47	-0.01%
5	1659.01	1657.864	-0.07%
6	2098.47	2116.923	0.88%
7	2419.68	2413.462	-0.26%

Mode Shapes

From Code: First six modes:



From ANSYS: First seven modes:



Another table:

Modes	RPM: 20000	Pre-twist: 15°	Stagger: 20°
	ANSYS (Hz)	RR	R_hub: 0.2m
		$n_s \times n_c$: 6×6	% Difference
1	464.7071	462.259	-0.53%
2	822.2366	821.832	-0.05%
3	1343.871	1346.649	0.21%
4	1705.789	1720.101	0.84%
5	1928.945	1934.76	0.30%
6	2453.493	2447.106	-0.26%
7	2646.897	2660.499	0.51%

5.5. Campbell Diagram and Speed-Dependent Behavior

[PLACEHOLDER: Figure 7 — Campbell diagram.] Caption: Natural frequencies of the lowest eight modes versus rotor speed from 0 to a maximum speed exceeding the operating range, with engine-order lines 1E to 6E. Identify the crossings of the first flap and first torsion modes with low engine orders, and mark the operating speed. Overlay the ANSYS results at discrete speeds as symbols. Discuss the different sensitivities of flap-dominated modes (strong stiffening) versus torsion- and edgewise-dominated modes (weak stiffening or softening) and relate them to the stress components entering $\boldsymbol{K}_\sigma$.

[PLACEHOLDER: Figure 8 — Decomposition of rotational effects.] Caption: Frequency shift of the first three modes versus speed with (a) $\boldsymbol{K}_\sigma$ only, (b) $\boldsymbol{K}_\sigma - \boldsymbol{K}_\Omega$, (c) $\boldsymbol{K}_\sigma - \boldsymbol{K}_\Omega$ with $\boldsymbol{G}$. Quantify the relative magnitudes of centrifugal stiffening, spin softening, and Coriolis coupling for a blade of this aspect ratio and stagger, and identify the speed at which the Coriolis-induced splitting of closely spaced modes becomes discernible.

[PLACEHOLDER: Figure 9 — Complex mode shapes and Coriolis coupling.] Caption: Real and imaginary parts of the mid-surface displacement for a pair of Coriolis-coupled modes at the operating speed, showing the phase lag

between flap and edgewise components that distinguishes a gyroscopic (travelling) mode from a standing mode. Report the modal phase collinearity index.

5.6. Dynamic Stability and Divergence Boundaries

[PLACEHOLDER: Figure 10 — Stability map.] Caption: Minimum eigenvalue $\lambda_{min}(\widetilde{\boldsymbol{K}})$ of the mass-normalised stiffened operator and maximum growth rate $maxRe\lambda$ versus rotor speed, extending to speeds at which λ_{min} becomes negative and the solver switches to the general branch. Identify the divergence speed and compare it with the speed at which the ANSYS prestressed modal analysis reports a zero or negative frequency. Note that, because the present model is linear about the reference geometry, the static-validity thresholds will typically be exceeded before divergence; discuss the implications for interpreting the predicted divergence speed as an upper bound on the linear-model validity rather than as a physical limit.

[PLACEHOLDER: Table 7 — Growing modes.] Caption: At the first speed for which unstable eigenvalues are detected: list each growing eigenvalue (real part in s^{-1}, imaginary part in Hz), its residual, and the dominant character of its mode shape.

5.7. Comparative Benchmark Validation against Published Literature

Li-Cheng paper

This is simplified/simple plate

Constant thickness(3mm) rectangular plate.

	Pre twist: 15°	offset: 0.5	RPM: 4k	Stagger: 45°	
Table 3	My RR	Li-Cheng Paper ANSYS	% Dev	Li-Cheng Paper	% Dev
	7.496	7.4951	0.01%	7.5559	-0.79%
	19.470	19.3556	0.59%	19.4017	0.35%
	23.116	23.0676	0.21%	23.298	-0.78%
	39.486	39.3061	0.46%	40.494	-2.49%
	45.890	45.6935	0.43%	45.5882	0.66%
		RMSE	0.397%		1.265%
	My RR	My ANSYS	% Dev		
	91.116	91.091	0.03%		
	236.654	236.481	0.07%		
	280.977	280.486	0.18%		
	479.962	472.819	1.51%		
	557.793	554.120	0.66%		
	629.034	621.176	1.26%		
	820.376	805.750	1.82%		
Parameter	ANSYS	Code-RR	% Diff		
Mass	1.42291519	1.41479059	-0.5710%		
Volume	0.00018126	0.00018023	-0.5699%		

RPM: 8k	**offset: 0.5**	**Stagger: 45°**		
My RR	**Li-Cheng Paper ANSYS**	**% Dev**	**Li-Cheng Paper**	**% Dev**

14.145	14.1175	0.19%	14.3692	-1.56%
25.406	25.2388	0.66%	25.2255	0.72%
38.400	38.2925	0.28%	39.2559	-2.18%
45.417	45.2517	0.37%	46.3747	-2.07%
58.239	57.9657	0.47%	56.874	2.40%
	RMSE	0.427%		1.883%
My RR	**My ANSYS**	**% Dev**		
171.939	171.682	0.15%		
308.811	308.025	0.26%		
466.759	464.781	0.43%		
552.046	545.985	1.11%		
707.898	703.099	0.68%		
871.709	865.160	0.76%		
969.147	953.755	1.61%		
	RMSE	0.86%		

5.8. Parametric Sensitivity Studies (Pre-twist, Taper, Camber, Hub Radius, Stagger)

[PLACEHOLDER: Figure 11 — Effect of pre-twist.] Caption: Lowest six frequencies at 0 and 4000 rpm versus total pre-twist $\theta_{tip} \in [-45°, 0°]$. Show the increasing flap–torsion coupling and the frequency veering between the second flap and first torsion branches.

[PLACEHOLDER: Figure 12 — Effect of taper and camber.] Caption: Frequencies versus (a) thickness taper $\tau_h \in [0,0.7]$, (b) chord taper $\tau_c \in [0,0.4]$, and (c) camber, comparing NACA 0012, 2412, 4412, and 6412 sections at fixed thickness. Discuss the opposing influence of camber on chordwise bending stiffness and torsional inertia.

[PLACEHOLDER: Figure 13 — Effect of hub radius and stagger.] Caption: Rotating frequencies at 4000 rpm versus hub radius $R_h/L \in [0,2]$ and root stagger $\theta_0 \in [-60°, 0°]$. Relate the hub-radius sensitivity to the radial stress level and the stagger sensitivity to the redistribution of the centrifugal load between the flap and edgewise directions and to the in-plane component of $\boldsymbol{K}_{\Omega}$.

5.9. 5.9 Computational Efficiency and DOF Reduction

[PLACEHOLDER: Table 8 — Computational effort.] Caption: Wall-clock time (single BLAS thread, specified CPU), peak memory, and degree-of-freedom count for the present method at Ritz orders 4×4, 6×6, 8×8, and 10×10, against the ANSYS shell and solid models of Tables 4 and 6 (including meshing, static prestress, and modal solve). Highlight the ratio of degrees of freedom required to achieve 0.5 % agreement on the first eight rotating frequencies.

6. Conclusions

A semi-analytical formulation has been developed for the free vibration and dynamic stability of rotating compressor blades represented as pre-twisted, tapered, cambered NACA four-digit airfoil shells. The formulation combines two elements that have hitherto been confined to separate methodological traditions. From the degenerated-continuum shell concept it takes the exact three-dimensional Jacobian mapping, so that the strain field is obtained from the true Cartesian displacement gradient without recourse to the Lamé parameters and curvature approximations of classical shell theories; from the Rayleigh–Ritz tradition it takes a global, L_2-orthonormal polynomial basis—scaled Jacobi functions in the span that enforce the clamped root exactly, and normalised Legendre functions in the chord—so that the entire blade is represented by a few hundred degrees of freedom without a finite element mesh.

The rotational physics is treated consistently in the blade-fixed frame. The centrifugal body force is integrated to a consistent load, the three-dimensional static equilibrium is solved, and the complete recovered stress tensor drives the initial-stress stiffness through the same kinematic operator and quadrature used for the dynamic operators. Coriolis coupling and spin softening enter through a skew-symmetric gyroscopic matrix and a centrifugal stiffness matrix that

include rotary inertia. A singularity-aware chordwise quadrature restores spectral integration accuracy in the presence of the square-root leading-edge behavior of the NACA thickness law and the curvature discontinuity of the camber line.

The quadratic gyroscopic eigenvalue problem is solved by an energy-scaled reformulation whose state matrix is exactly real skew-symmetric when the stiffened operator is positive definite. The Hermitian eigenproblem that results returns purely imaginary eigenvalues, so that natural frequencies are free of the spurious real parts that contaminate generalized solvers on ill-scaled companion matrices and that would otherwise be misinterpreted as damping or growth. When positive-definiteness is lost, a time-scaled general solve preserves the growth rates that identify divergence. Every static and dynamic solution is accompanied by residual and validity diagnostics, permitting the range of applicability of the linear reference-state prestress approximation to be delimited objectively.

The verification and validation program of Section 5 establishes the convergence of the Ritz expansion and quadrature, the agreement with shell- and solid-element finite element solutions at rest and under rotation, and the ability of the method to produce Campbell diagrams and stability maps at a fraction of the finite element cost. The framework extends directly to laminated composite sections by replacing the constitutive matrix, to arbitrary section families by replacing the camber and thickness functions, and to nonlinear prestress by iterating the equilibrium solve on the deformed geometry, none of which requires modification of the kinematic or discretization machinery.

7. Data Availability Statement

8. Conflict of Interest

9. Acknowledgements

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